

Project Title

Deep learning based Single Shot EPI DWI image distortion correction and Reconstruction

Clinical and Research Motivation

Single-shot EPI diffusion-weighted imaging (ss-EPI DWI) is widely used in routine brain MRI and is clinically important for detecting stroke, tumours, and other brain pathologies. However, ss-EPI DWI is commonly affected by geometric distortion, especially near the skull base, temporal lobes, and regions with strong susceptibility artefacts. Current distortion correction approaches often depend on hardware-based solutions, high-performance gradients, vendor-specific software licenses (e.g. Siemens RESOLVE), field-mapping techniques, or additional reverse phase-encoding acquisitions. These methods may increase scan time or require scanner capabilities that are not available in low-resource clinical settings. This limitation is relevant in low- and middle-income countries, where many centres use entry-level 1.5T or older 3T MRI systems and may not have access to advanced readout-segmented diffusion sequences due to cost, licensing, or vendor limitations. High clinical workload also makes longer “distortion-free” sequences difficult to include in routine protocols. An AI-based tool could help improve the anatomical fidelity of routine ss-EPI DWI images, particularly at the skull base and temporal lobes, and may reduce the risk of misinterpretation of strokes, tumours, or other pathology in resource-limited settings.

Research Questions

- How well do existing state-of-the-art models perform on a locally acquired dataset?
- Can foundation models improve the performance of current state-of-the-art and pretrained models?
- Can a qualitative evaluation protocol be developed for datasets that lack ground-truth labels?

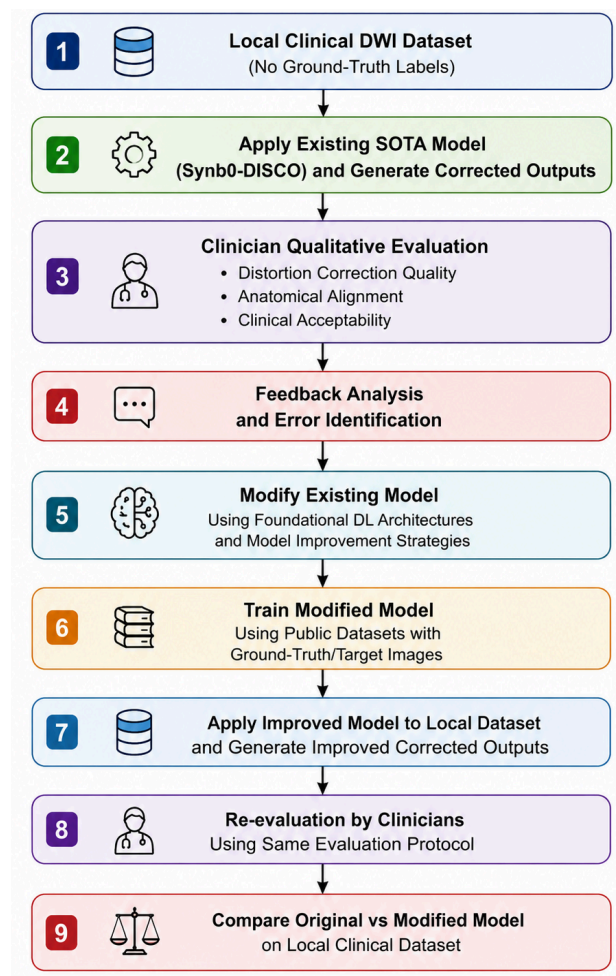
Objectives

- To evaluate the performance of existing state-of-the-art models on a locally acquired dataset.
- To investigate whether foundation models can enhance the performance of existing state-of-the-art and pretrained models.
- To develop a qualitative evaluation protocol for datasets that do not have ground-truth labels.

Proposed Methodology

First, the locally available dataset will be tested using an existing state-of-the-art DWI distortion correction model (Synb0-DISCO in our case). Since the local dataset does not contain ground-truth undistorted labels, a qualitative evaluation protocol will be developed for clinicians to assess the model outputs on these data. The clinician evaluation will provide feedback on distortion correction quality, anatomical alignment, and clinical acceptability.

Based on this feedback, the existing SOTA model will be modified using foundational deep learning architectures and model improvement strategies. The modified model will then be trained using publicly available datasets that contain ground-truth or correction-derived target images. After training, the improved model will again be applied to the local dataset. The corrected outputs will be re-evaluated by clinicians using the developed evaluation protocol, allowing comparison between the original SOTA model and the modified model on the same local clinical dataset.



Dataset and Label Structure

The training data will be based on the Human Connectome Project dataset. HCP T1-weighted images were acquired using a 3D MPAGE sequence with TE = 2.1 ms, TR = 2400 ms, slice thickness = 0.7 mm, flip angle = 8° , FOV = 224×224 mm, and voxel size = 0.7×0.7 mm. HCP diffusion images were acquired using a single-shot EPI sequence with b-values of 1000, 2000, and 3000 s/mm², with 90 diffusion directions and 6 b = 0 s/mm² images per shell. The diffusion acquisition used TE = 89.5 ms, TR = 5520 ms, slice thickness = 1.25 mm, flip angle = 78° , FOV = 210×180 mm, and 1.25 mm isotropic voxel size. For training, the model input will include distorted b0/DWI images and corresponding T1-weighted images, while the target label will be the corrected or undistorted b0 image generated from opposite phase-encoded diffusion acquisitions.

The local dataset was acquired from Jeebanta Advanced Kathmandu Imaging Pvt. Ltd. and consists of paired T1-weighted and diffusion-weighted MRI scans from 289 subjects. All required ethical documentation and approval requirements were completed before using the data, and the data were used in de-identified form. The T1-weighted images were acquired using a 3D T1-weighted TFE sequence with TE = 3.6/3.7 ms, TR = 7.5/7.6 ms, and flip angle = 8° . The T1-weighted images have a matrix size of $213 \times 320 \times 320$, voxel resolution of $0.8 \times 0.72 \times 0.72$ mm, and LAS orientation.

The local diffusion-weighted images were acquired using a single-shot spin-echo EPI sequence with TE = 87 ms, TR = 3723 ms, flip angle = 90° , and phase-encoding/fold-over direction along A-P. The diffusion images have a 4D matrix size of $192 \times 192 \times 29 \times 5$, voxel resolution of $1.2 \times 1.2 \times 5$ mm, and LAS orientation. The diffusion acquisition consists of five volumes, including one b = 0 s/mm² volume and four diffusion-weighted volumes acquired at b = 1000 s/mm². Unlike HCP, this local dataset does not have ground-truth undistorted b0 labels, so it will be used for inference and clinician-based qualitative evaluation.

Chosen Model or Baseline Workflow

The deep learning model used in the current SOTA model 3D U-Net. So it will be used for inferencing. The model takes two input channels, a T1-weighted image and a distorted b0 image and synthesizes an undistorted b0 image. The synthesized undistorted b0 will then be merged with the distorted b0 and will go through FSL TOPUP for final distortion correction.

Component	Description
Baseline deep learning model	3D U-Net
Input	T1-weighted image + distorted b0 image
Output	Synthesized undistorted b0 image
Post-processing	Synthesized b0 + distorted b0 passed through FSL TOPUP
Proposed modification	Replace the 3D U-Net backbone using foundational model like SwinUNETR
Training data	Publicly available HCP dataset with correction-derived undistorted b0 targets
Local data use	Inference, clinician evaluation

Novelty for the selected method

The novelty of this project is doing clinical-based evaluation on an existing state-of-the-art distortion correction workflow on a locally acquired clinical dataset and then modifying the model according to clinicians' feedback. Since the local dataset does not contain ground-truth undistorted b0 labels, the project also introduces a clinician-based qualitative evaluation protocol. After the baseline evaluation, the current 3D U-Net-based approach will be improved using a foundational model architecture like SwinUNETR.

Expected training parameters and optimization approach

The original SOTA model used a 3D U-Net with two input channels and one output channel, using leaky ReLU, instance normalization, 5-fold cross-validation, and MSE loss. It was trained for 100 epochs with the Adam optimizer, learning rate 0.0001, betas 0.9/0.999, and weight decay 1e-5.

In this project, we will initially follow these SOTA training parameters and modify them only if needed during training of the new SwinUNETR-based model.

Anticipated Implementation Challenges

The main implementation challenge is that the training pipeline involves time-consuming preprocessing, including registration between T1-weighted and DWI/b0 images, normalization, transformation to template space, and later transformation back to subject space. These steps can make both training and inference slow.